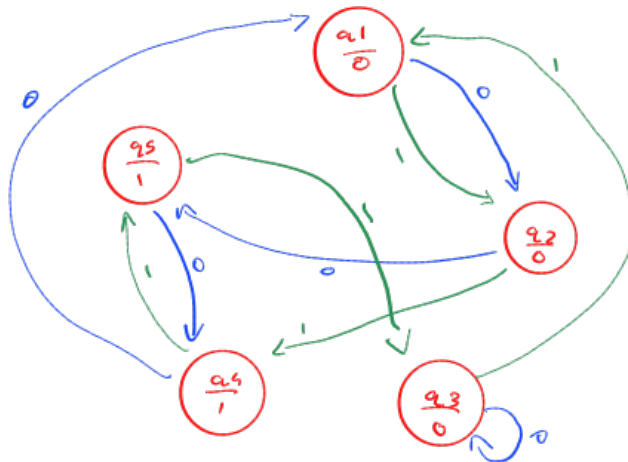


Conversion of State Diagram and Table with Input Sequence

2. Moore Machine

Draw state diagram and apply input sequence "0 0 0" to q4 and determine output sequence



	0	1
q1	q2	q2
q2	q5	q4
q3	q3	q1
q4	q1	q5
q5	q4	q3

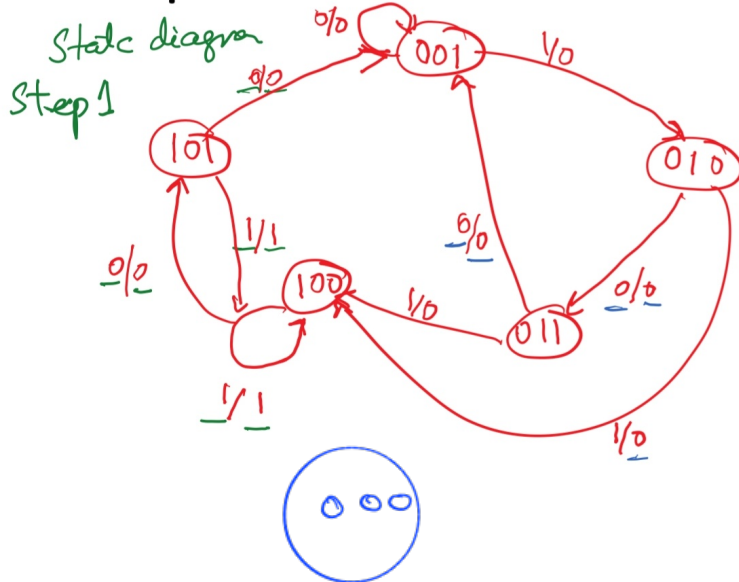
IS = 000 → 04

OS = 1001

Example of Unused states

$n=3$ bits

$$2^n = 2^3 = 8$$



Step 2

State table

Pr State	$x=0$	Nxt state $x=1$	y
001	001	010	0
010	011	100	0
011	001	100	0
100	101	100	0
101	001	100	0

Unused 000, 110, 111 } don't care

000
001
010
011
100
101
110
111

001

Example of Unused states

q1	q2	q3	x	Q1 +	Q2 +	Q3 +	j1	k1	j2	k2	j3	k3	y
0	0	0	0	0	0	0	X	X	X	X	X	X	
0	0	0	1	0	1	0	X	X	X	X	X	X	
0	0	1	0	0	1	1	0	X	1	X	X	0	0
0	0	1	1	1	0	0	1	X	0	X	X	1	0
0	1	0	0	0	1	1	0	X	X	0	1	X	0
0	1	0	1	1	0	0			X	1	0	X	0
0	1	1	0	0	0	1							0
0	1	1	1	1	0	0							0
1	0	0	0	1	0	1							0
1	0	0	1	1	0	0							1
1	0	1	0	0	0	1							0
1	0	1	1	1	0	0							1

Excitation of Jk

→ 00	0 X
→ 01	1 X
10	X 1
11	X 0

step 4 - kmap & eqn
steps - Design

1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

110
111

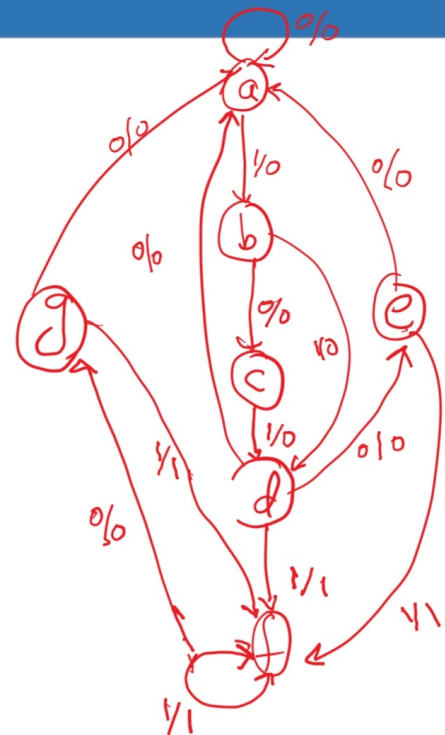
PS	NS		y	
	n=0	n=1	m=0	m=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	d	0	1
e	a	d	0	1

e=g

d=f

Example of State Reduction

Pk State	Next State		Y	
	x=0	x=1	x=0	x=1
a	a	b	0	0
b	c	d	0	0
c	a	d	0	0
d	e	f d	0	1
e	a	f d	0	1
f	g e	f	0	1
g	a	f	0	1



Example of State Reduction

State Assignment

a-000

b-001

c-010

d-011

e-100

Pk-St	N-S		O/P(Y)	
	x=0	x=1	x=0	x=1
<u>000=a</u>	<u>000 a</u>	001 b	0	0
<u>001=b</u>	010 c	011 d	0	0
<u>010=c</u>	<u>000 a</u>	011	0	0
<u>011=d</u>	100 e	011	0	1
<u>100=e</u>	000 a	011	0	1

Example of State Reduction

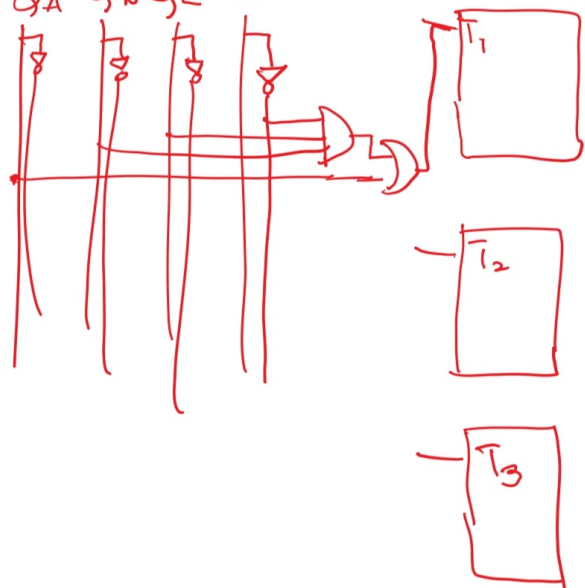
Excitation of T

ps $Q_A Q_B Q_C$	i/p (x)	Nxt $Q_A Q_B Q_C$	o/p (y)	T1	T2	T3
<u>000</u> = a	0	<u>000</u> a	0	0	0	0
<u>000</u> = a	1	<u>001</u> b	0	0	0	1
<u>001</u> = b	0	<u>010</u> c	0	0	1	1
<u>001</u> = b	1	<u>011</u> d	0	0	1	0
<u>010</u> = c	0	<u>000</u> a	0	0	1	0
<u>010</u> = c	1	<u>011</u> d	0	0	0	1
<u>011</u> = d	0	<u>100</u> e	0	1	1	1
<u>011</u> = d	1	<u>011</u> d	1	0	0	0
<u>100</u> = e	0	<u>000</u> a	0	1	0	0
<u>100</u> = e	1	<u>011</u> d	1	1	1	1

00	0
01	1
10	1
11	0

Example of State Reduction

$Q_A Q_B Q_C x$



T_1 $Q_C x$

$Q_A Q_B$	00	01	11	10
00				
01				1
11	X	X	X	X
10	1	1	X	X

$$T_1 = Q_A + Q_B Q_C \bar{x}$$

Miguel

Example of State Reduction

ps	i/p (x)	Nxt	o/p (y)	T1	T2	T3
<u>000</u> = a	0	<u>000</u> = a	0	0	0	0
<u>000</u> = a	1	<u>001</u> = b	0	0	0	1
<u>001</u> = b	0	<u>010</u> = c	0	0	1	1
<u>001</u> = b	1	<u>011</u> = d	0	0	1	0
<u>010</u> = c	0	<u>000</u> = a	0	0	1	0
<u>010</u> = c	1	<u>011</u> = d	0	0	0	1
<u>011</u> = d	0	<u>100</u> = e	0	1	1	1
<u>011</u> = d	1	<u>011</u> = d	1	0	0	0
<u>100</u> = e	0	<u>000</u> = a	0	1	0	0
<u>100</u> = e	1	<u>011</u> = d	1	1	1	1

Excitation of T

00	0
01	1
10	1
11	0

P.S $\propto$ N.S

you sent a
line bro